The Setting

We work in a compact, riemanian, orientable, smooth Manifold (M, g) of dimension m together with a Morse-Smale funktion

$$f : M \to \mathbb{R} \tag{1}$$

We assume that q is a critical point of index $k+1$ and p is a critical point of index k. we define $f(q) = b$ and $f(p) = a$ and assume that there is no kritical point in $f^{-1}(a,b) \subseteq M$. We define the sub and superlevelsets

$$M^t := \{x \in M | f(x) \leq t\} \quad \text{and} \quad M_t := \{x \in M | f(x) \geq t\}\,, \tag{2}$$

and the constants

$$c \in (a,b) \quad , \quad \varepsilon > 0 \text{ small} \quad , \quad T > 0 \text{ big}\,. \tag{3}$$

With this we define the sets:

$$N_q := \{x \in M_c | f(\varphi_{-T}(x)) \leq b + \varepsilon\}\,, \tag{4}$$
$$L_q := \{x \in N_q | f(x) = c\}\,, \tag{5}$$
$$N_p := \{x \in M^c | f(\varphi_T(x)) \geq a - \varepsilon\}\,, \tag{6}$$
$$L_p := \{x \in N_p | f(\varphi_T(x)) = a - \varepsilon\}\,, \tag{7}$$

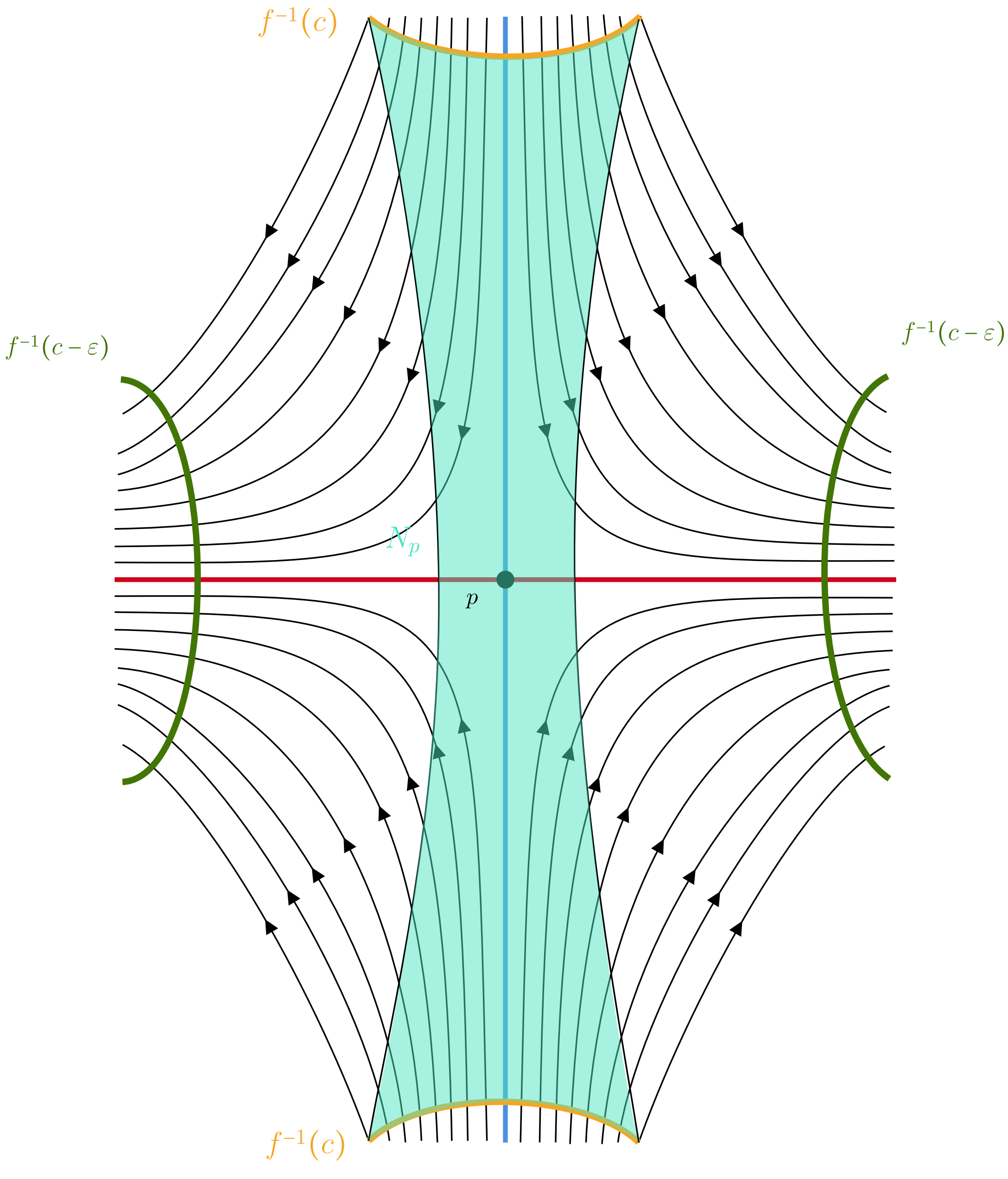
and finally:

$$C := N_p \cup N_q\,, \tag{8}$$
$$B := N_p \cup L_q\,, \tag{9}$$
$$A := L_p \cup (L_q \overset{\circ}{-} N_p)\,. \tag{10}$$

Lemma 0.0.1. *We claim that*

1. (N_q, L_q) *is a regular index pair for* q .
2. (C, B) *is an index pair for* q .
3. (N_p, L_p) *is a regular index pair for* p .
4. (B, A) *is an index pair for* p .
5. N_p *is a tubular neighbourhood of* $W(\to p) \cap M^c$.

Proof. The statements 1-4 are easy to proof by the definitions. The regularity can be derived from a general fact for pairs (X, Y) in metric spaces: If Y is closed in X and there is a neighbourhood of Y that is open in X such that Y is a strong deformation retract of U. So the only thing left to show is, that N_p is a tubular neighbourhood. in [**MorseTheorySalmbon**] in the proof of lemma 3.2 Salamon claims this to be true without a proof (page 119 in the attached source). Similar, in [**banyaga2004lectures**] Banyaga claims this (also without any argument). I find this difficult to proof, since we cannot use the flow for the map from the normal bundle, as points leave N_p along the flow: The Set N_p is sketched in the figure below.



So to proof this, we first formulate two lemmas. First we restate/ refine the λ-lemma:

Lemma 0.0.2. *(λ−lemma) Let M be ansmooth manifold and let p be a hyperbolic fixpoint of a diffeomorphism φ on M with $\dim W(p \to) = r$ for $0 < r < m = \dim M$. Let N be an injective immersed submanifold of M that is invariant under φ and has a point of transverse intersection q with $W(\to p)$. Then for any r−cell neighbourhood D of q in N and any $r-cell$ neighbourhood B of p in $W(p \to))$ there is a smaller cell $\tilde{D} \subset D$ and an $n \in \mathbb{N}$ such that B is ε C^1-close to $\varphi^m(D)$ for all $m > n$*

Proof. In **??** we proofed the same with the only difference, that we defined the r−cell D arbitrary. We now want to show that this r−cell can be defined to be the image of a moved smaller cell of a given $r - cell$. So lets inspect the already given proof:

We started by moving the point of transverse intersection q closer to the fixpoint by applying the diffeomorphism to it. Then we defined an neighbourhood D_0 of that moved point in the stable manifold (compare **??**). Here we were able to get a hold of the inclination. Then we moved D_0 via applying the isomorphism closer to the fixed point and restricted it to the dimension of the unstable manifold. This restriction we can refine as follows: First we define D_{n^2} to be the moved D_0. Lets say we moved q n−times. In other words $q \in \varphi^{-n}(D_{n^2})$ Then there exist a small r−cell neighbourhood of q that lives in $D \cap \varphi^{-n}(D_{n^2})$. That is because $\varphi^{-n}(D_{n^2})$ is an in N open neighbourhood of y, hence $\varphi^{-n}(D_{n^2}) \cap D$ is open in D. Hence we can find a ball around y in D that is contained in the intersection. We call that r−cell $\tilde{D}$. Then $\varphi^n(\tilde{D})$ is an r−dimensional subset of D_{n^2} and the rest of the proof can be done with that subset. All arguments work, as that subset is transverse to $W(\to p)$. □

Next we want to see, that locally the flow is monoton in the normal direction:

Lemma 0.0.3. *Let U be a normal neighbourhood of $W(\to p)$. For $x \in W(\to p)$ we define U_x to be the image of the normal fibre above x. Then for any $T > 0$ big enough and $\varepsilon > 0$ small enough, the function $f \circ \varphi_T : U_x \to R$ is monoton decreasing in a small ball of radius ε meaning that if $\lambda > 1, v \in U_x$ with $\|v\| < \|\lambda v\| < \varepsilon$ then $f \circ \varphi_T(v) < f \circ \varphi_T(\lambda v)$.*

Proof. This is an application of the λ-lemma. □

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